MATH 122 — Calculus II · Exam 2 Solutions

Part I: Integration by Parts

1. Evaluate $\displaystyle\int x^{20}\ln x\,dx$

LIATE: $\ln x$ is **L**ogarithmic and x^{20} is **A**lgebraic, so $u=\ln x$.

$u=\ln x,\ dv=x^{20}\,dx \Rightarrow du=\frac{1}{x}\,dx,\ v=\frac{x^{21}}{21}$

$$\int x^{20}\ln x\,dx=\frac{x^{21}}{21}\ln x-\int\frac{x^{21}}{21}\cdot\frac{1}{x}\,dx=\frac{x^{21}}{21}\ln x-\frac{1}{21}\int x^{20}\,dx$$

$$\boxed{\int x^{20}\ln x\,dx=\frac{x^{21}}{21}\ln x-\frac{x^{21}}{441}+C}$$

2. Evaluate $\displaystyle\int 3x^3e^x\,dx$

Pull out the constant, then use the **tabular method** with $u=x^3$, $dv=e^x\,dx$.

Sign	D (differentiate x^3)	I (integrate e^x)
$+$	x^3	e^x
$-$	$3x^2$	e^x
$+$	$6x$	e^x
$-$	6	e^x
	0	e^x

Multiply each signed D entry by the I entry one row below and add:

$$\int x^3e^x\,dx=x^3e^x-3x^2e^x+6xe^x-6e^x+C$$

$$\boxed{\int 3x^3e^x\,dx=3e^x\left(x^3-3x^2+6x-6\right)+C}$$

3. Evaluate $\displaystyle\int \cot^{-1}x\,dx$

LIATE: write the integrand as $\cot^{-1}x\cdot 1$; $\cot^{-1}x$ is **I**nverse trig, so $u=\cot^{-1}x$, $dv=dx$.

$u=\cot^{-1}x,\ dv=dx \Rightarrow du=-\frac{1}{1+x^2}\,dx,\ v=x$

$$\int \cot^{-1}x\,dx=x\cot^{-1}x+\int\frac{x}{1+x^2}\,dx$$

Let $w=1+x^2$, $dw=2x\,dx$:

$$\int\frac{x}{1+x^2}\,dx=\frac{1}{2}\int\frac{dw}{w}=\frac{1}{2}\ln(1+x^2)$$

(No absolute value is needed since $1+x^2>0$.)

$$\boxed{\int \cot^{-1} x\,dx = x\cot^{-1} x + \frac{1}{2}\ln(1+x^2) + C}$$

4. Evaluate $\displaystyle\int_0^{\pi} x\sin x\,dx$

LIATE: x is **A**lgebraic and $\sin x$ is **T**rigonometric, so $u = x$.

$u = x,\ dv = \sin x\,dx \Rightarrow du = dx,\ v = -\cos x$

$$\int_0^{\pi} x\sin x\,dx = \big[-x\cos x\big]_0^{\pi} + \int_0^{\pi}\cos x\,dx = \big[-x\cos x + \sin x\big]_0^{\pi}$$
$$= (-\pi\cos\pi + \sin\pi) - (0 + \sin 0) = (\pi + 0) - 0$$

$$\boxed{\int_0^{\pi} x\sin x\,dx = \pi}$$

Part II: Trigonometric Integrals

5. Evaluate $\displaystyle\int \sin^7 x\,dx$

Use the **reduction formula for sine**:

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x\,\cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx$$

$n = 7$**:** $\displaystyle\int \sin^7 x\,dx = -\frac{1}{7}\sin^6 x\cos x + \frac{6}{7}\int \sin^5 x\,dx$

$n = 5$**:** $\displaystyle\int \sin^5 x\,dx = -\frac{1}{5}\sin^4 x\cos x + \frac{4}{5}\int \sin^3 x\,dx$

$n = 3$**:** $\displaystyle\int \sin^3 x\,dx = -\frac{1}{3}\sin^2 x\cos x + \frac{2}{3}\int \sin x\,dx = -\frac{1}{3}\sin^2 x\cos x - \frac{2}{3}\cos x$

Substitute back:

$$\int \sin^7 x\,dx = -\frac{1}{7}\sin^6 x\cos x + \frac{6}{7}\left[-\frac{1}{5}\sin^4 x\cos x + \frac{4}{5}\left(-\frac{1}{3}\sin^2 x\cos x - \frac{2}{3}\cos x\right)\right]$$

$$\boxed{\int \sin^7 x\,dx = -\frac{1}{7}\sin^6 x\cos x - \frac{6}{35}\sin^4 x\cos x - \frac{8}{35}\sin^2 x\cos x - \frac{16}{35}\cos x + C}$$

Alternative (odd power): write $\sin^7 x = (1-\cos^2 x)^3\sin x$ and let $u = \cos x$, $du = -\sin x\,dx$:

$$\int \sin^7 x\,dx = -\int (1-u^2)^3\,du = -\cos x + \cos^3 x - \frac{3}{5}\cos^5 x + \frac{1}{7}\cos^7 x + C$$

Both answers are correct; they are equal as functions.

6. Evaluate $\displaystyle\int \sin^3 x\cos^2 x\,dx$

This is $\int \sin^m x\cos^n x\,dx$ with $m = 3$ and $n = 2$. Since m is odd, follow the **"m odd"** row: split off a factor of $\sin x$, use $\sin^2 x = 1 - \cos^2 x$, and substitute $u = \cos x$.

$$\int \sin^3 x\cos^2 x\,dx = \int (1-\cos^2 x)\cos^2 x\,\sin x\,dx$$

With $u = \cos x$, $du = -\sin x\,dx$:

$$= -\int (1-u^2)u^2\,du = -\int (u^2 - u^4)\,du = -\frac{u^3}{3} + \frac{u^5}{5} + C$$

$$\boxed{\int \sin^3 x \cos^2 x\,dx = -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C}$$

7. Evaluate $\displaystyle\int \sin 10x \cos 3x\,dx$

Use $\sin A \cos B = \frac{1}{2}\big[\sin(A-B) + \sin(A+B)\big]$ with $A = 10x$, $B = 3x$:

$$\sin 10x \cos 3x = \frac{1}{2}\big[\sin 7x + \sin 13x\big]$$

$$\int \sin 10x \cos 3x\,dx = \frac{1}{2}\left(-\frac{\cos 7x}{7} - \frac{\cos 13x}{13}\right) + C$$

$$\boxed{\int \sin 10x \cos 3x\,dx = -\frac{\cos 7x}{14} - \frac{\cos 13x}{26} + C}$$

8. Evaluate $\displaystyle\int 4 \sin 3x \cos 5x \sin 7x\,dx$

Step 1: $\sin 3x \cos 5x = \frac{1}{2}\big[\sin(-2x) + \sin 8x\big] = \frac{1}{2}\big[\sin 8x - \sin 2x\big]$, so $4 \sin 3x \cos 5x = 2 \sin 8x - 2 \sin 2x$.

Step 2: multiply by $\sin 7x$:

$$4 \sin 3x \cos 5x \sin 7x = 2 \sin 8x \sin 7x - 2 \sin 2x \sin 7x$$

Step 3: use $\sin A \sin B = \frac{1}{2}\big[\cos(A-B) - \cos(A+B)\big]$:

- $2 \sin 8x \sin 7x = \cos x - \cos 15x$
- $2 \sin 2x \sin 7x = \cos(-5x) - \cos 9x = \cos 5x - \cos 9x$

So the integrand is $\cos x - \cos 5x + \cos 9x - \cos 15x$.

Step 4: integrate term by term:

$$\boxed{\int 4 \sin 3x \cos 5x \sin 7x\,dx = \sin x - \frac{\sin 5x}{5} + \frac{\sin 9x}{9} - \frac{\sin 15x}{15} + C}$$

Part III: Partial Fractions

9. Evaluate $\displaystyle\int \frac{11}{(2x-5)(x+3)}\,dx$

$$\frac{11}{(2x-5)(x+3)} = \frac{A}{2x-5} + \frac{B}{x+3} \quad \Rightarrow \quad 11 = A(x+3) + B(2x-5)$$

Match coefficients: $11 = (A+2B)\,x + (3A-5B)$, so $A + 2B = 0$ and $3A - 5B = 11$, giving $A = 2$, $B = -1$.

Check: $x = -3$ gives $11 = -11B$; $x = \frac{5}{2}$ gives $11 = \frac{11}{2}A$.

$$\int \left(\frac{2}{2x-5} - \frac{1}{x+3}\right) dx = 2 \cdot \frac{1}{2} \ln|2x-5| - \ln|x+3| + C$$

(The $\frac{1}{2}$ comes from the substitution $w = 2x - 5$, $dw = 2\,dx$.)

$$\boxed{\int \frac{11}{(2x-5)(x+3)}\,dx = \ln|2x-5| - \ln|x+3| + C = \ln\left|\frac{2x-5}{x+3}\right| + C}$$

10. Evaluate $\displaystyle\int \frac{4}{x^2(x-2)}\,dx$

$$\frac{4}{x^2(x-2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} \quad \Rightarrow \quad 4 = Ax(x-2) + B(x-2) + Cx^2$$

Match coefficients: $4 = (A+C)\,x^2 + (-2A+B)\,x - 2B$, so

$$\begin{cases} A + C = 0 \\ -2A + B = 0 \\ -2B = 4 \end{cases} \quad \Rightarrow \quad B = -2,\ A = -1,\ C = 1$$

Check: $x = 0$ gives $4 = -2B$; $x = 2$ gives $4 = 4C$.

$$\int \left(-\frac{1}{x} - \frac{2}{x^2} + \frac{1}{x-2}\right) dx$$

$$\boxed{\int \frac{4}{x^2(x-2)}\,dx = -\ln|x| + \frac{2}{x} + \ln|x-2| + C}$$